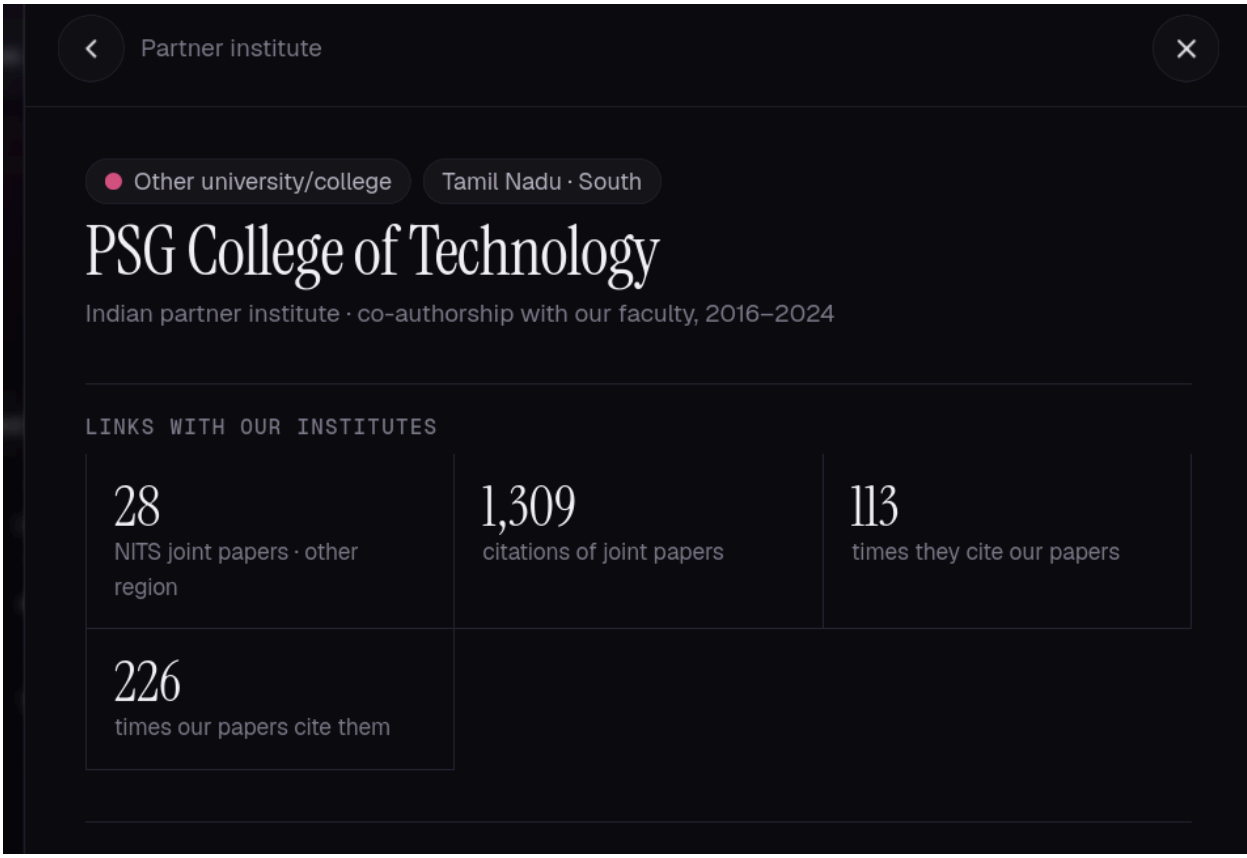
- We built dataset and analysed for 3 places MANIT Bhopal, NIT jamshedpur, NIT silchar
- **Key takeaways**
    - **Most number of papers from NIT-Silchar- why?**
    - **Manit highest H index- why**
    - **Nits highest q1 share- what is q1 share and why again**
    - **We computed how many papers came out every year, whose lowest whose highest and why**
    - **Where all did we fetch out data from, what were the verification methods**
- **Publications**
    - **Annual output has been grouped per year and per conference**
    - **Q1 per year, what's the analysis**
    - **0 conference papers on coreA*, core A, need to check**
    - **Research areas- most and least**
    - **Field impact what does the bubble size mean**
    - **Have listed top journals and top conferences, but need to list it per field**



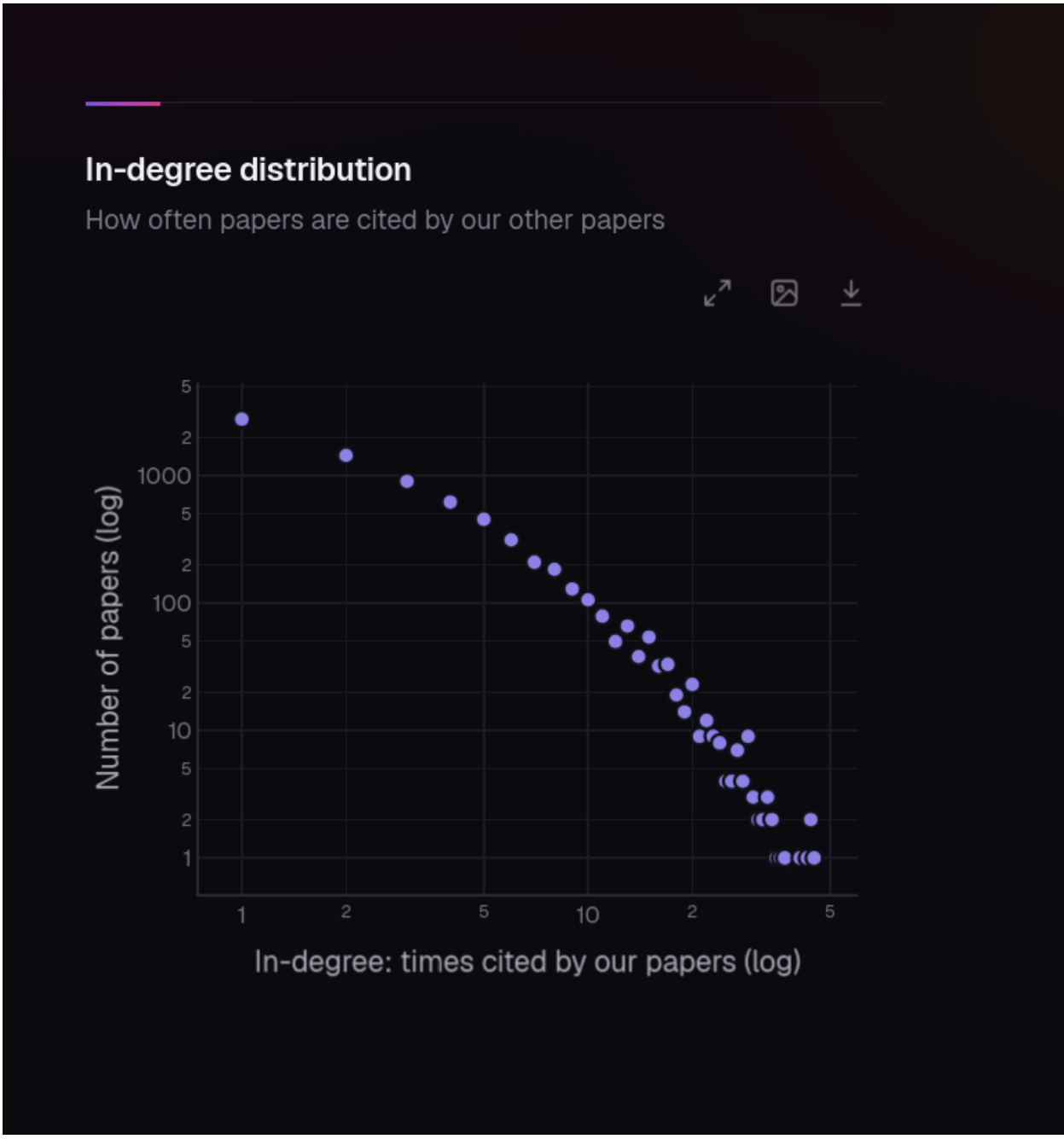
    - **Difference between computer science and computer applications**
    - **Check unknown departments- seems sus**



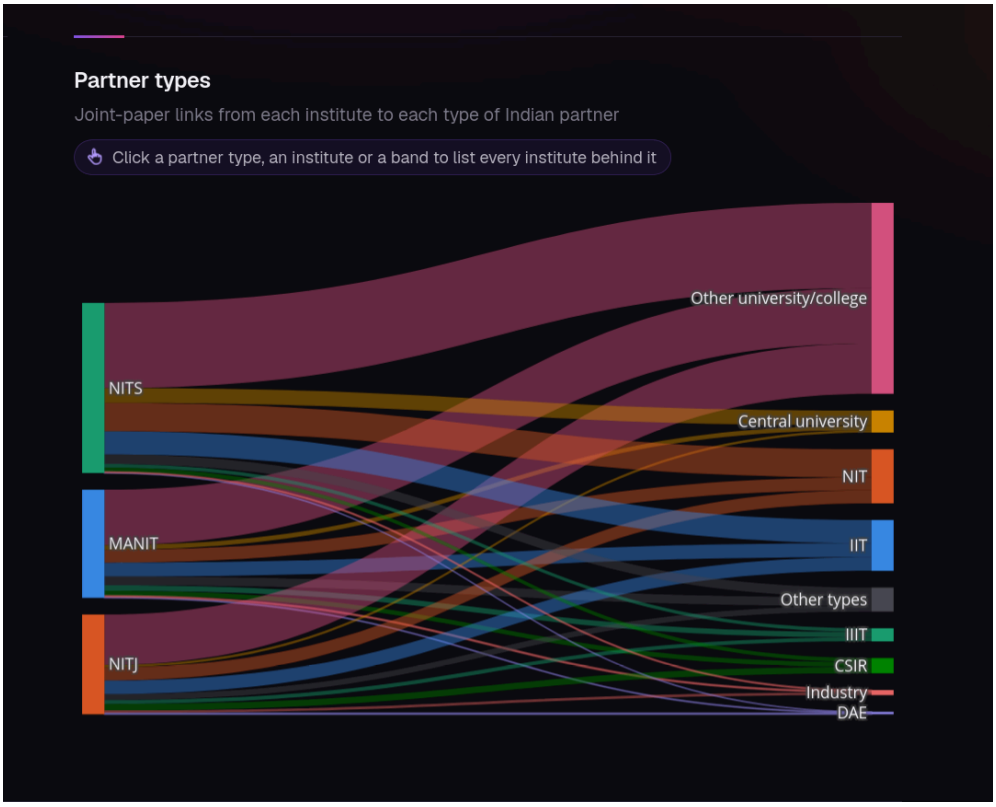
- **Who is they here**
- **How is h index computed**
- **Collaboration tab needs checking**
- **Faculty analysis- each prof most used conference, highest conference or journal submitted to**
- **Check proceeding series**
- **In citations tab, cites is not there for all (why did we only restrict it to corpus)**



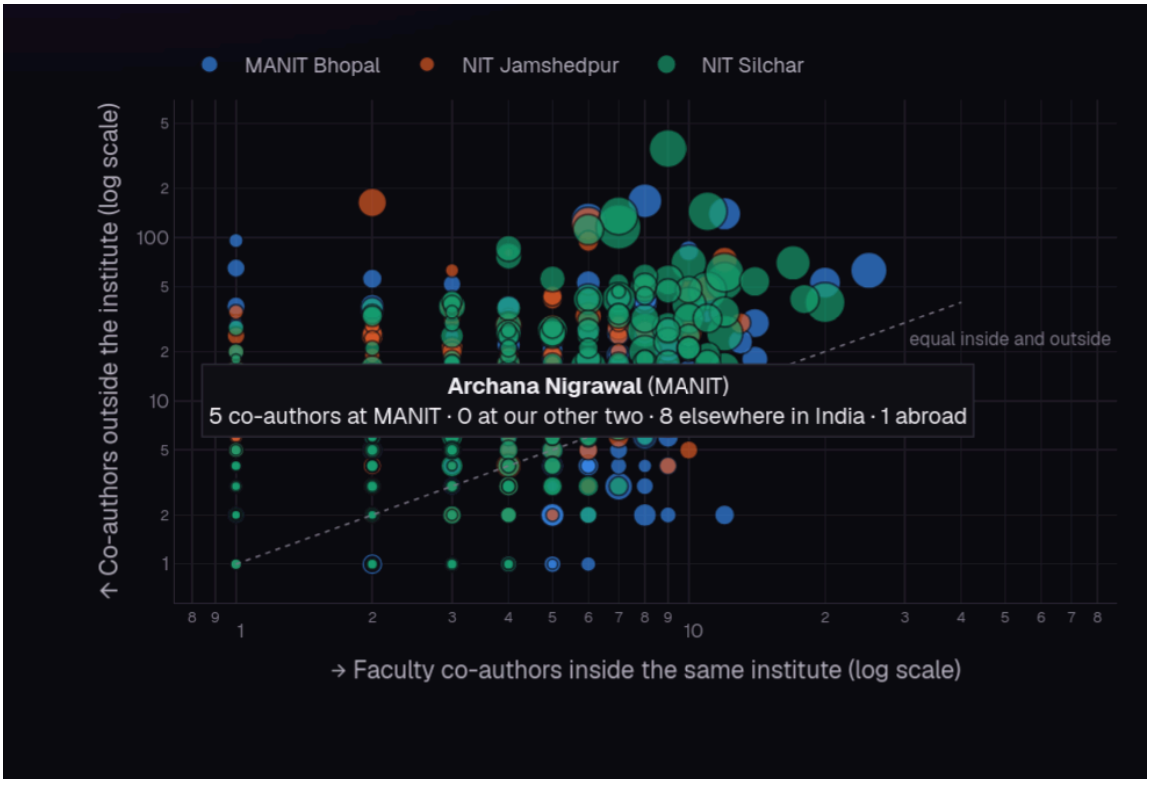
- **Understand this graph**



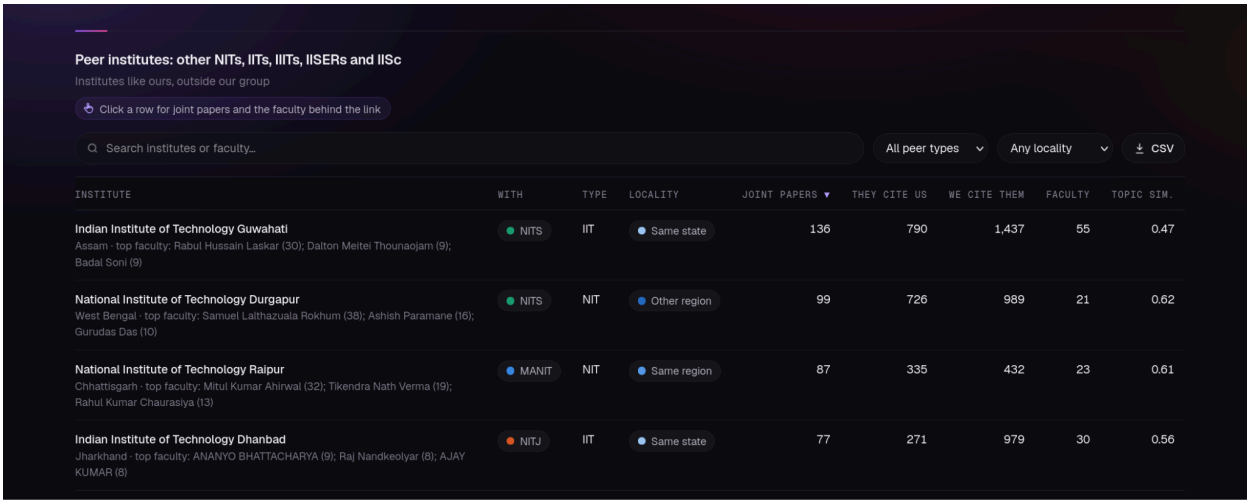
- **We need one for outdegree dist as well**
- **Reach tab: how many collaboration institutes were considered**
- **Shared indian partner institute, strongest faculty pair across**



- **Collab with iiit hyderabad- not there in this graph**

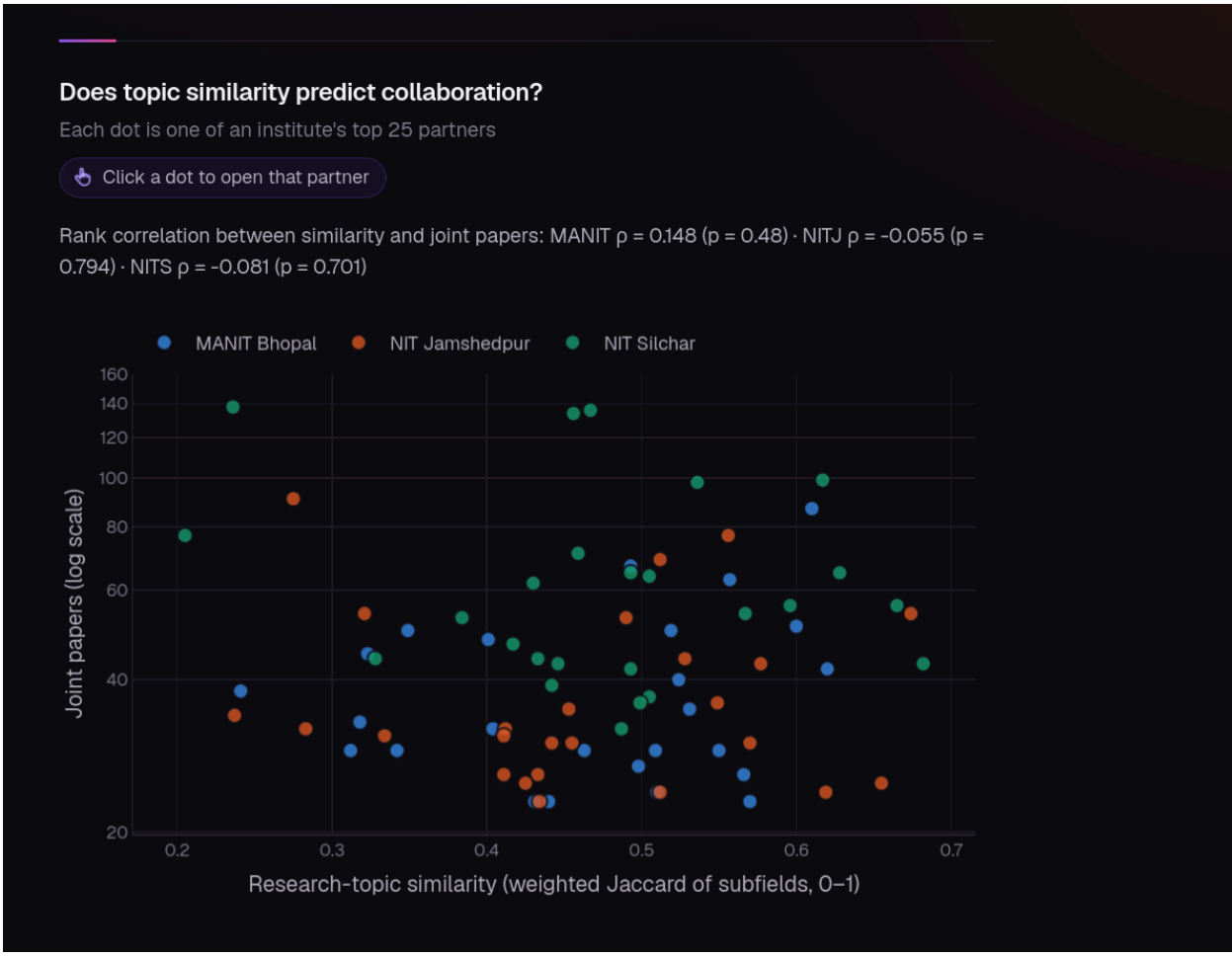


- **Where citations come from**

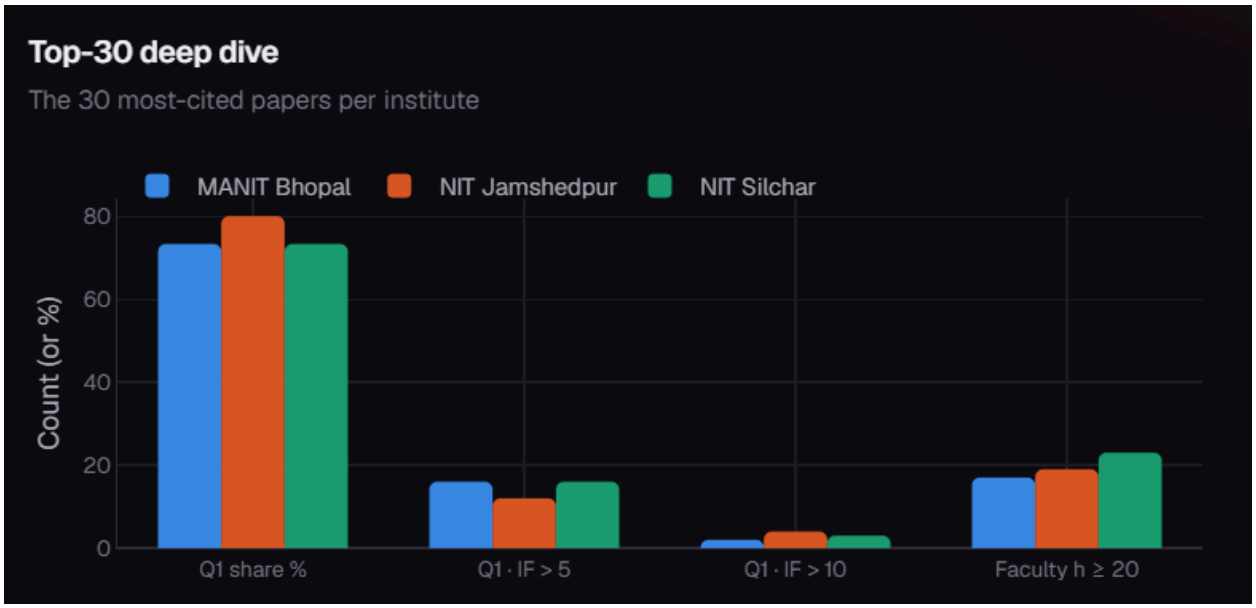
- 

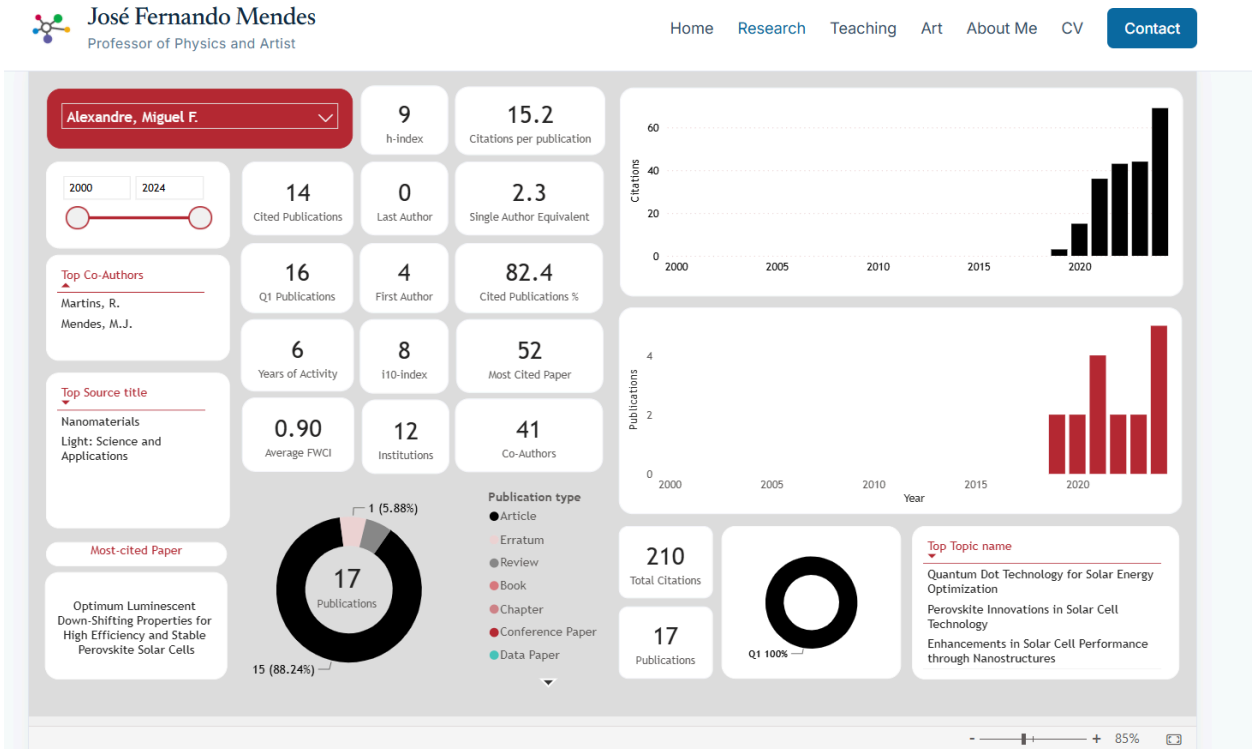
- **What is the final column**
- 

- **Abroad**


Does topic similarity predict collaboration?
Each dot is one of an institute's top 25 partners
Click a dot to open that partner
Rank correlation between similarity and joint papers: MANIT ρ = 0.148 (p = 0.48) · NITJ ρ = -0.055 (p = 0.794) · NITS ρ = -0.081 (p = 0.701)
MANIT Bhopal
NIT Jamshedpur
NIT Silchar
Joint papers (log scale)
160
140
120
100
80
60
40
20
0.2
0.3
0.4
0.5
0.6
0.7
Research-topic similarity (weighted Jaccard of subfields, 0–1)

-
- **Meaning**
-


Top-30 deep dive
The 30 most-cited papers per institute
MANIT Bhopal
NIT Jamshedpur
NIT Silchar
Count (or %)
80
60
40
20
0
Q1 share %
Q1 · IF > 5
Q1 · IF > 10
Faculty h ≥ 20


José Fernando Mendes
Professor of Physics and Artist
Home
Research
Teaching
Art
About Me
CV
Contact
Alexandre, Miguel F.
9
h-index
15.2
Citations per publication
2000
2024
14
Cited Publications
0
Last Author
2.3
Single Author Equivalent
16
Q1 Publications
4
First Author
82.4
Cited Publications %
6
Years of Activity
8
i10-Index
52
Most Cited Paper
0.90
Average FWCI
12
Institutions
41
Co-Authors
Top Co-Authors
Martins, R.
Mendes, M.J.
Top Source title
Nanomaterials
Light: Science and Applications
Most-cited Paper
Optimum Luminescent Down-Shifting Properties for High Efficiency and Stable Perovskite Solar Cells
1 (5.88%)
17
Publications
15 (88.24%)
Publication type
Article
Erratum
Review
Book
Chapter
Conference Paper
Data Paper
Citations
60
40
20
0
2000
2005
2010
2015
2020
Publications
4
2
0
Year
210
Total Citations
17
Publications
Q1 100%
Top Topic name
Quantum Dot Technology for Solar Energy Optimization
Perovskite Innovations in Solar Cell Technology
Enhancements in Solar Cell Performance through Nanostructures
85%

PAPER	INST.	CITES	DOI
ZnO nanostructured materials and their potential applications: progress, challenges and perspectives Nanoscale Advances · 2022	NITS	1,104	DOI ↗
Hydrogen embrittlement in different materials: A review International Journal of Hydrogen Energy · 2018	MANIT	971	DOI ↗
Per- and polyfluoroalkyl substances in water and wastewater: A critical review of their global occurrence and distribution The Science of The Total Environment · 2021	NITJ	930	DOI ↗
Floating photovoltaic power plant: A review Renewable and Sustainable Energy Reviews · 2016	MANIT	878	DOI ↗
Solar–wind hybrid renewable energy system: A review Renewable and Sustainable Energy Reviews · 2016	MANIT	843	DOI ↗
Green synthesis of silver nanoparticles using plant extracts and their antimicrobial activities: a review of recent literature RSC Advances · 2021	NITS	802	DOI ↗
Analysis of k-Fold Cross-Validation over Hold-Out Validation on Colossal Datasets for Quality Classification IEEE International Conference on Advanced Computing (IACC) · 2016	MANIT	709	DOI ↗
Sustainalism: An Integrated Socio-Economic-Environmental Model to Address Sustainable Development and Sustainability Sustainability · 2023	MANIT	703	DOI ↗
The role and significance of Magnesium in modern day	NITJ	699	DOI

LENS All three MANIT NITJ

Paper explorer

17,906 papers · 259,892 citations

Search or sort, then click a row for details

Search titles… | Any type | Any rank | All fields | Any impact | Any team size | Any Indian links | CSV

PAPER	INST.	YEAR	RANK	CITES ▾	PAGERANK	
ZnO nanostructured materials and their potential applications: progress, challenges and perspectives Nanoscale Advances · Materials Science	NITS	2022	Q1	1,104	80.80	DOI ↗
Hydrogen embrittlement in different materials: A review International Journal of Hydrogen Energy · Materials Science	MANIT	2018	Q1	971	181.20	DOI ↗
Per- and polyfluoroalkyl substances in water and wastewater: A critical review of their global occurrence and distribution The Science of The Total Environment · Environmental Science	NITJ	2021	Q1 IF>10	930	32.00	DOI ↗
Floating photovoltaic power plant: A review Renewable and Sustainable Energy Reviews · Environmental Science	MANIT	2016	Q1 IF>5	878	279.40	DOI ↗
Solar–wind hybrid renewable energy system: A review Renewable and Sustainable Energy Reviews · Energy	MANIT	2016	Q1 IF>5	843	144.40	DOI ↗
Green synthesis of silver nanoparticles using plant extracts and their antimicrobial activities: a review of recent literature RSC Advances · Materials Science	NITS	2021	Q2	802	50.40	DOI ↗
Analysis of k-Fold Cross-Validation over Hold-Out Valid… IEEE International Conference on Advanced Computing (IACC) · Computer Science · conference			Q4	709	64.60	DOI ↗

LENS: All three | MANIT | NITJ | NITS — YEARS: 2016 to 2026

Papers with more than 10 authors

Large-team papers compared with the rest

INSTITUTE	GROUP	PAPERS	SHARE	MEAN CITES	MEDIAN CITES	SHARE OF CITES	Q1 %
MANIT	more than 10 authors	22	0.38%	21.27	8.5	0.52%	78.9
MANIT	10 or fewer authors	5,781	99.62%	15.6	4.0	99.48%	33.1
NITJ	more than 10 authors	14	0.39%	45.07	30.0	1.43%	57.1
NITJ	10 or fewer authors	3,620	99.61%	12.02	3.0	98.57%	26.5
NITS	more than 10 authors	34	0.4%	26.71	12.0	0.72%	64.5
NITS	10 or fewer authors	8,479	99.6%	14.76	5.0	99.28%	36.5



	MANIT	NITJ	NITS
Total citations	13,526	8,356	9,813
Median citations	367	203.5	272.5
Citations of #30	275	156	191
Journal papers	27	30	30
Q1 papers	22	24	22
Median venue h-index	261	128	243.5
Median best-author h	21	38	29

Problems, most serious first:

1. **PageRank and In-degree mostly measure self-citation, not influence.** 87.6% of the citation links inside the corpus go from a faculty member to their own earlier paper, and 96% stay within one institute. The paper with the highest PageRank has 77 citations in total, and all 22 of the ones from inside the corpus are self-citations. Ranking by PageRank and ranking by citations agree only weakly (Spearman 0.55), and the two top-100 lists share just 10 papers. Presenting these columns as influence measures, as the explorer does now, is misleading.
2. **About 210 journal papers are labelled "conference".** The cause is build_tables.py:60, which treats any venue with "proceedings" in its name as a conference. This catches PNAS, *Proceedings of the Royal Society A*, the whole *Proc. IMechE Part A–O* family and *Proc. ICE*. As a result, these papers appear under the Conference filter and drop out of journal-only figures such as Q1 share.
3. **About 111 titles appear twice.** Most are the same paper registered under two DOIs. Inderscience issues a temporary `10.1504/…10000xxx` DOI as well as the final one,

and *Resource-Efficient Technologies* has two DOI prefixes. These papers are counted twice and their citations are split between the copies.

4. **86 papers have no DOI.** The citation network is joined on DOI, so these papers always show PageRank 0 and In-degree 0, even when they are cited.
5. **Q1 share is calculated two different ways.** The quick stats at the top of the explorer count journals only. The Top 30 summary uses all 30 papers: MANIT shows 22/30 = 73.3%, but its journal-only figure would be 22/27 = 81.5%. Separately, the Quartile filter includes 32 conference papers labelled Q1 (proceedings also get Scopus quartiles).

- Which institute is doing the best
- All detailed info about each faculty
- More info on the institute
- Check why no core A* and core A
- In collaborations what r each of the terms- eigenvector, betweeness,
- 86 papers dont have DOI - so page rank is 0
- **Q1 share is calculated two different ways.** The quick stats at the top of the explorer count journals only. The Top 30 summary uses all 30 papers: MANIT shows 22/30 = 73.3%, but its journal-only figure would be 22/27 = 81.5%. Separately, the Quartile filter includes 32 conference papers labelled Q1 (proceedings also get Scopus quartiles).
-

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* faculty, papers ~~citns~~, Median H-index Q1 share be innovative
findings, betweenness, quartile dist.
paper by no. of authors
filters

Publication
o/p by yr & type
top journals
10 - 20 most cited journals.
Faculty → h-index distri (2016-2026).
faculty performance → overall h & btw 2016-2026
degree & btwness.
Q1
Q1 IF > 10
Q1 IF > 5.
#authors < 10
authors > 10.
pre-prints.

performance of institute & faculty shld be shown.
show many links btw 2 faculty
Explain key findings through network analysis.
Use ML analysis. Use GDP, climate analysis.
statistics & network
provide author links for each

Citations Tab:

1. The colour grid is flipped (the most serious one)
- The page claims to show who cites whom, but it shows the reverse.
- Any conclusion drawn from it is backwards. For example, it shows NITJ citing MANIT more, when really MANIT cites NITJ more (135 vs 118).

2. The grid uses raw counts, so the biggest institute looks most important
- NITS has 17,363 links within itself and MANIT 6,091. That's mainly because NITS has far more papers (8,513 vs 5,803), not because NITS cites itself more.
- **The fair way to show it** is as percentages: what share of each institute's citations go to its own papers.
  - MANIT: 94.3%
  - NITJ: 91.9%
  - NITS: 97.5%
- These are all similar: all three institutes mostly cite their own work and rarely each other.

3. The PageRank ranking favours old papers
- Of the top 50 papers by PageRank, 24 are from 2016, and none are from after 2022.
- Yet 2023–2026 has the most papers (over 2,000 a year).
- **Why:** older papers have had more years to collect citations. A 2025 paper can't have been cited much yet.
- So the "most influential" list really means "oldest well-cited". The professor may spot this.
- **Fix:** rank within each year, or add a "best per year" view.

4. Joint papers are assigned to an institute arbitrarily
- If a paper has authors from both NITJ and NITS, the grid gives it only to whichever institute is listed first.
- That depends on author order, so the counts shift slightly. The table counts the paper for both institutes instead.
- **Fix:** count it for both in the grid too, or show joint papers as their own group.

>>> Add the thing that A star papers are from students and no faculty included in them

PAPERS tab

New insights from the paper data

1. **The internal citation network shows research groups building on their own work, not influence between the institutes.** NIT Silchar has the highest self-citation share (90%). Its top in-corpus figure, Md. Ahmaruzzaman, receives 1,837 citations from inside the corpus, 97% of them self-citations. Only 44 papers out of 17,906 involve more than one of the three institutes. This makes a strong point for the report.
2. **Each institute leans on a few people.** Citations are highly concentrated: the Gini coefficient is 0.77 for MANIT, 0.75 for Jamshedpur and 0.70 for Silchar. At Jamshedpur, two faculty account for about 25% of all citations: Sunil Kumar 14.6% and Balram Ambade 10.6%. At MANIT, Kumarasamy Sudhakar alone accounts for 12.9%. Silchar is the most broadly spread: its top 1% of papers hold 14% of its citations, against about 21% at the other two.
3. **MANIT's citations rely heavily on review articles.** Papers with "review" in the title are 3.0% of MANIT's output but bring 17.2% of its citations, and they make up 14 of its top 30. The other two institutes get under 10% of their citations from reviews.
4. **Jamshedpur's top 30 is largely one 2020 surge.** 14 of its top 30 papers are from 2020 and 15 of the 30 involve Sunil Kumar. These are fractional-calculus models of epidemics, tumours and predator–prey systems, which OpenAlex files partly under "Medicine".
5. **Quartile strongly predicts citations, and conferences fall far behind.** For journal papers published up to 2023, Q1 papers have median citations of about 27–32, while Q4 papers have about 2–3, roughly a tenfold gap. Conference papers have median citations of 2–4, and 15–22% have never been cited. Most of them appear in Springer *Lecture Notes* series and *Materials Today: Proceedings*.
6. **Journal quality is rising but conference volume is not falling.** The Q1 share of journal papers grew from about 27% in 2016–17 to about 38% in 2024–26. Meanwhile the conference share stayed at roughly 35–40% every year.
7. **A few fields punch above their size.** Compared with their share of all papers, Agricultural & Biological Sciences appears 4.1× as often among the top 1% most-cited, Mathematics 3.9× and Energy 3.4×. Materials Science is the largest field but only 1.4×.
8. **Recent papers worth watching, ranked by citations per year.** I used this ranking to correct for paper age:
    - "Sustainalism" (MANIT, 2023): about 210 citations per year.
    - Silchar's AI smart-farming reviews: about 120 per year.
    - A multimodal sentiment analysis survey in *ACM Computing Surveys* (Silchar): about 110 per year.

- Citations: only if the cited papers are in corpus its present, else shows 0.
-